OUR LADY OF LOURDES COLLEGE
5031 GEN. T. DE LEON, VALENZUELA CITY

# The Effectiveness of Portable Water Purifier for the Drinkable Water of Senior High School Students in Our Lady of Lourdes College

**A Quantitative Research Paper**

**Presented to the**

**Senior High School Department of**

**Our Lady of Lourdes College**

**In Partial Fulfillment of the Subject**

**Practical Research 2**

**STEM 12 - 1**

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# CHAPTER I

# THE PROBLEM AND ITS BACKGROUND

## Introduction

Water is a fundamental necessity for life, and ensuring access to safe drinking water is essential for maintaining good health, particularly in educational environments. Schools have a responsibility to provide students with clean and uncontaminated water sources. However, water quality in schools can be compromised by various contaminants, including bacteria, viruses, heavy metals, and chemicals. These contaminants pose potential health risks to students, potentially impacting their healthy lifestyle.

According to Mamuad, Roselle Y. (2022), Portable, low-cost water filtration studies have been conducted to address the issue of providing affordable and reliable access to clean drinking water. These studies focus on developing filtration systems that are effective, easily produced, and sustainable. One study developed a low-cost dispenser-type water filtration system made of clay, which significantly improved the quality of tap water and met the standards for drinking water safety.

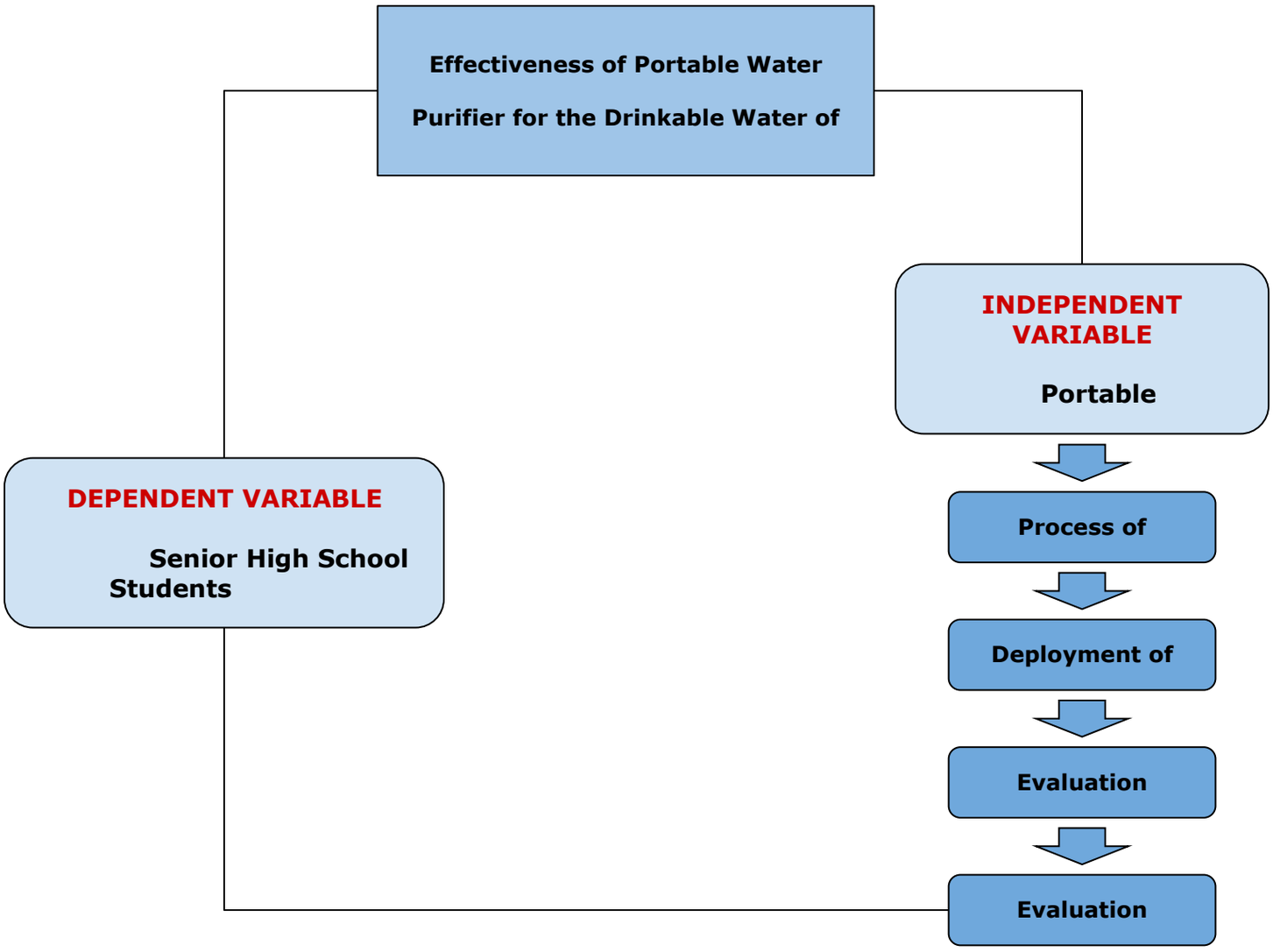
According to Joseph Lovegrove (2019), he proposed a portable water filtration device using hydrophilic capillary fiber membranes, which filled a significant portion of the housing and effectively removed contaminants from water. Furthermore, According to Ajaybhaskar V. et al. (2019), it is necessary to develop a low-cost water purifier using locally available filter materials, such as rice husk and sand, to provide pure drinking water to families, students, and the community.

Portable water purifiers offer a potential solution to address school water quality concerns. This quantitative study aims to investigate the effectiveness of portable water purifiers in improving the drinking water quality of senior high school students at Our Lady of Lourdes College. The study will specifically focus on measuring the reduction of crucial contaminants – bacteria, heavy metals, and chemicals – before and after purification. Additionally, the research will explore the potential impact of portable water purifiers on student perceptions of water safety and their drinking water consumption habits.

The findings of this study will provide valuable insights into the feasibility and effectiveness of portable water purifiers to enhance water safety within a school setting. The results have the potential to inform the decision-making of individuals as well as to provide additional information to the existing body of knowledge regarding this topic; it may also serve as a basis for determining whether the Philippines may use this information to develop suggestions regarding the water quality improvement strategies at Our Lady of Lourdes College and contribute to broader discussions on ensuring safe drinking water in schools.

**Conceptual Framework**

The purpose of this study is to make a small apparatus that purifies water at Our Lady of Lourdes College. It is meant to give clean water that won't cost anything. This apparatus cleans the water so it is safe to drink. Our goal is to help Senior High School Students to stay healthy by providing clean and drinkable water.



**Theoretical Framework**

Rosenstock, I. M. developed **The Health Belief Model** (1974), which offers a theoretical framework for comprehending how people perceive and behave in relation to health interventions. It aids in the analysis of how educators, staff members, and students view the value of adopting water filtration systems and clean drinking water.

This research is guided by the **Health Promotion Model (HPM)**, which posits that individual, environmental, and behavioral factors influence health outcomes. By providing students with access to portable water purifiers, this study aims to encourage their use as a healthy behavior, ultimately improving their water consumption and overall health.

The **Theory of Hygiene and Sanitation,** advanced by Wasike (2010), emphasizes cleanliness and no germs. This also extends to the organizations that permit hygiene to be successful. This theory suggests that it includes interested people and organizations in the community, including political leaders, faith-based organizations, and non-governmental organizations. If there should be meaning to hygiene and sanitation, they should be casting the net wide. Despite polluted water being a major cause of infectious illnesses, health is affected through the spread of organic and inorganic chemicals that are dangerous to health and include chlorinated solvents that result in cancer, trihalomethanes that are toxic to the liver and kidney, heavy metals such as lead damaging nerve and brain and congenital disabilities and polychlorinated biphenyls causes liver damage, and may also lead to

cancer. Additionally, the provision of safe drinking water and the effective removal of bodily waste by suppliers and food handlers are essential for human health and well-being.

**Environmental Health Perspective** by Sharma Et Al. (2017), the knowledge of drinking water quality, including health hygiene and sanitation aspects, among students and community people is an urgent need. The environmental health perspective on drinking water is a critical topic, especially when educating students about the quality and health impacts of drinking water. Lack of awareness and education among students and communities regarding drinking water quality can lead to significant health issues. It is essential to address this gap through educational initiatives that highlight the importance of safe drinking water and the potential health risks associated with contaminated water sources. By providing students with knowledge about the health impacts of drinking water quality, we can empower them to make informed choices and advocate for sustainable water management practices that protect both human health and the environment. This education is vital in fostering a culture of responsibility towards water resources and promoting behaviors that contribute to a healthier and more sustainable future.

**Statement of the Problem**

The objective of this research is to evaluate the effectiveness of a new portable water purifier. The primary focus is to assess the performance and usability of improving water quality within the school premises. Moreover, this study also aims to investigate the following:

1. What is the Demographic Profile of Respondents
    a. Name
    b. Age
    c. Sex
    d. Grade Level
2. What is the pH level of the Portable Water Purifier in removing contaminants from the water?
3. Does the Use of Portable Water Significantly Reduces Bacteria in Drinking Water?
4. What is the perception of the students about the Water Quality of Portable Water Purifier?
5. Is there a significant difference between the quality of the Water Before and After Treatment?

**Significance of the Study**

The study's consequences for the most significant recipient are explained in this section. The following are the beneficiaries of the study:

First are **Students**; ensuring that the students have access to clean and safe drinking water is crucial for their overall health. Knowing how effective portable water purifiers are will help to guarantee that students always have access to safe drinking water, lower their chance of contracting illnesses from contaminated water, and improve their general health.

Secondly, **Teachers** In order for the students to succeed academically, teachers are essential. Teachers may focus on their teaching duties with confidence, presents topics in biology, chemistry, environmental science, and water purification in a practical setting. Promoting critical thinking abilities through practical tests and data analysis on water quality and purification techniques. Knowing that their kids won't be negatively impacted by water quality problems when they know that portable water purifiers efficiently supply drinkable water.

The **Parents** Particularly when their kids are away at school, parents worry about the health and safety of their younger ones. Parents might be concerned about the possible health hazards linked with contaminated water when they know that their children have access to clean drinking water while at school.

The **School Administrators** will likely monitor, knowing that the students on the school premises are safe from any possible health hazards. They will be at ease, and they will also gain knowledge about the potential dangers of contaminated water. They can implement rules and regulations to improve the school policy.

The **Doctors** were in charge, and we are truly aware of how crucial clean water is to preserving health. Doctors can be informed about possible actions to achieve better health outcomes among students by lowering the risk of waterborne infections and by knowing how effective portable water purifiers are.

**Future Researchers** are responsible for Future investigations in comparable situations or involving different populations. They can draw on the findings of research on the efficacy of portable water purifiers in educational environments. It provides information about the viability and effectiveness of such initiatives, directing the course of future studies targeted at enhancing public health and water quality.

**Hypothesis**

The use of portable water purifier will significantly improve the quality of drinkable water available to senior high school students, leading to a reduction in waterborne illnesses and an overall enhancement in student health and well-being.

**Ho:** There is no significant difference between the Quality of Drinking Water between the Before and After Treatment.

**Ha:** There is a significant difference.

**Scope and Delimitations**

This study aims to investigate the water quality within OUR LADY OF LOURDES located in 5031 Gen. T. De Leon to determine if there is a significant difference in the water quality. Additionally, we will gather feedback from 50 Senior High School Students about their experience with our product.

**Definition of Terms**

This section introduces specific terms that hold significance within the research context, thereby establishing a shared understanding for readers.

**Chlorine Level -** The amount of chlorine presents in water, often added as a disinfectant to kill bacteria.

**Contaminants -** Impurities or unwanted substances found in water, such as bacteria, viruses, heavy metals, or chemicals.

**pH Level -** A measure of the acidity or alkalinity of water. A pH of 7 is considered neutral, while lower values indicate acidity, and higher values indicate alkalinity.

**Portable Water Purifier -** A device that can be easily transported and used to remove contaminants from water, making it safe to drink.

**Total Dissolved Solids (TDS) -** The total amount of dissolved inorganic and organic substances in water**.**

**Turbidity -** A measure of the cloudiness of water caused by suspended particles.

**Water Quality -** A measure of the physical, chemical, and biological characteristics of water that determine its suitability for a particular purpose, such as drinking.

# CHAPTER II

# REVIEW RELATED LITERATURE AND STUDIES

## Review Related Literature

According to the World Health Organization (WHO) 2023, around 2 billion people worldwide do not have access to safely managed drinking water services. This leads to devastating consequences, with approximately 485,000 deaths annually due to diarrheal diseases caused by inadequate sanitation and water contamination. Portable water purifiers offer a promising solution to this global issue. By effectively removing harmful contaminants and pathogens from water sources, they can transform dirty or contaminated water into safe drinking water in an instant.

In the review of related literature of Ateneo de Manila University (2021), it stated that Water is necessary. The human body consists of approximately 60% water. The majority of individuals consume a minimum of 8 glasses of water every day, taking into account their daily activities, geographical region, and overall health. This is done in order to regulate body temperature, facilitate joint movement, and eliminate waste products from the body. Ateneo recognizes the need to ensure that drinking water is readily available to the busy individuals who visit the university. The objective is to offer complimentary potable water in various areas inside the campus for the use of students, employees, and visitors alike. Drinking fountains are carefully positioned adjacent to toilets and in every cafeteria on campus to promote water intake throughout the day. The Office for Food Safety and Quality Assurance

(OFSQA) is primarily responsible for ensuring that Ateneo de Manila University complies with both local and global food safety, sanitation, and nutrition regulatory standards. This is done to prioritize the well-being of the Ateneo community. This involves the surveillance and establishment of criteria to enhance the safety, excellence, and nutritional value of food.

The investigation conducted by Ahmed et al. (2022) involved the preparation of activated charcoal with a substantial surface area of 4320–3800 m2/g through the chemical and thermal processing of walnut wastes. This activated charcoal exhibited notable adsorption capabilities. Various ratios of potassium hydroxide (KOH) were used to sonicate Iraqi walnut shells. The combination was subsequently subjected to calcination at various temperatures using an electric oven until the optimal thermal conditions for achieving a significantly large activated surface area and enhanced performance were determined. The activated charcoal obtained was subsequently refined to eliminate any remaining potassium hydroxide (KOH) and metallic impurities. The investigations uncovered that the quality of the manufactured activated charcoal was on par with or exceeded that of commercially available counterparts in terms of both physical and adsorption qualities. It was characterized by its ability to degrade methylene blue and remove heavy components during the process of water purification. In addition, Alves et al. (2021) examine the application of biological activated carbon in water treatment and its associated methodologies, aiming to assess the efficacy of Activated Carbon (AC) in purifying water for human consumption. To achieve this objective, the study attempts to determine

the methodologies employed in the manufacture of AC, its applications, and the advantages it offers. Additionally, it will explore the potential of producing AC from vegetable waste for water treatment purposes. Activated carbon possesses the capacity to selectively gather gases, liquids, and impurities within its pores. This attribute explains its extensive application in filtration systems and water treatment. As an adsorbent, coal effectively retains specific types of impurities in its pores, such as large particles that contribute to undesirable color, taste, or odor in water. Coal is utilized in the treatment of effluents for the purposes of clarity, deodorization, and purification. And yet both, indicate that activated carbon, a form of charcoal, is highly effective in water purification. It has a large surface area, which allows it to effectively absorb contaminants such as heavy metals, organic compounds, and impurities that cause color, taste, or odor issues in water. And in the study of Rosario Ph.D. (2012), Water is the most abundant compound on earth, and is essential for life. Unfortunately, it is becoming more difficult to provide potable water to the rapidly expanding human population due to depletion of water sources and increasing pollution. Nevertheless, the Philippines is committed to achieving the United Nations Millennium Development Goal Number 7 to halve by 2015 the proportion of the population without sustainable access to safe drinking water and safe sanitation. According to the National Statistical Information Center, 84.1% of the country's population as of 2008, or about 77 million had attained sustainable access to safe drinking water and improved sanitation and the target is to reach 86.5% by 2015.

The findings of Punzalan C. H (2020), suggest that creating an environmental education program that will not only sustain but also enhance the level of awareness and extent of practice of senior high school students in relation to environmental problems and the principles of sustainable development. One way to achieve this is by incorporating elective courses focused on environmental education into the senior high school curriculum, particularly for students in the Science, Technology, Engineering, and Mathematics (STEM) strand, as they are already familiar with the related learning objectives. In order to effectively educate students about these issues, it is recommended to promote environmental stewardship within the school premises through the implementation of policies such as zero waste practice, bring our own bottle, and waste segregation program. This will ensure that the message reaches beyond the classroom. The school administrators and instructors are also urged to expand the classroom settings to include outdoor activities and programs, such as participating in tree or mangrove planting activities and coastal clean-ups and proper use of Water Purification. In order to evaluate whether certain legislations, including those related to education, environment, and local government units, are effective, it is necessary to undertake a comprehensive policy assessment.

According to Wu, J., Cao, M., Tong, D., et al. (2021), they stated that the use of filtration stone beads for water filtration has been studied for their ability to remove impurities and improve the taste and quality

of water. Mineral stone beads, also known as remineralization filters, are used in water filtration systems to restore healthy minerals to water that has been treated with reverse osmosis (RO) filtration. These filters are designed to adjust the alkalinity of RO-filtered water by reintroducing calcium and magnesium ions, which are essential for maintaining the pH balance and preventing corrosion of pipes and appliance, It also have shown that mineral stone beads are effective in improving the taste and quality of RO-filtered water, making it more palatable and non-corrosive. However, there are also concerns about the marketing claims made by some manufacturers regarding the health benefits of these filters, and there is no scientific evidence to support these claims. Despite this, mineral stone beads are popular following RO filtration systems and are used in various applications for water filtration.

In the study of Dhuria, A. (2023), He stated that Silica sand is a versatile material widely used for water filtration due to its effectiveness in removing impurities and contaminants from water. It has been a popular choice for water filtration systems for decades, and it is known for its ability to trap dirt, debris, organic material, chemical compounds, and metals in water, thereby cleaning and purifying them. Silica sand's high porosity allows it to absorb large amounts of water while remaining effective, making it suitable for filtering large volumes of water quickly. One of the key advantages of using silica sand for water filtration is its natural composition, making it safe for human and animal consumption as it is non-toxic and does not produce harmful byproducts. Additionally,

silica sand is cost-effective compared to other filters like carbon filters, making it an affordable option for water filtration systems. Its durability and resistance to extreme temperatures, pressures, and bacteria make it ideal for use in water filtration systems exposed to harsh conditions, ensuring long-lasting and efficient filtration performance.

According to the review of Dutta, S. (2022), the pH level of filtered water depends on the type of filter used. For example, activated carbon filters can lower the pH level of water, while reverse osmosis filters do not have a significant effect on the pH level. In general, filtered water has a slightly lower pH level than unfiltered water, but the difference is usually not significant. The pH level of filtered water is typically within the range of 6.5 to 7.5, which is considered safe for drinking. Furthermore, the pH level of water is an important factor to consider when using a portable water purifier. While some filters are designed to raise the pH level of water, it is important to ensure that the water remains within the optimal pH range of 6 to 8.5 to ensure its safety and quality. Regular testing and maintenance of the water filter help ensure that the pH level remains within this range and that the water is safe to drink.

According to Ottenhall et al. (2018), they found that by giving the surfaces of cellulose fibers in two types of commercial paper filters a positive net charge, they were able to trap bacteria through electrostatic interactions without releasing any biocides. The filters were normal as just paper filtration in single layers or multilayers, along with the structure inside of the Water Purifier. The results showed that all filters,

using both types of filter papers and a number of layers, could physically remove more than 99.9% of E. coli from water and that the 3 L modified filters could remove more than 97% of cultivatable bacteria from natural water samples. Thus, the researchers tried another water-based process at room temperature to alter the surfaces of cellulose fibers in paper filters. The modified filters were able to physically remove more than 99.9% of E. coli from water and more than 97% of cultivatable bacteria from natural water samples. And they both seem very effective. This study suggests that paper filtration can be an effective and affordable method for water purification, particularly for point-of-use water purification systems. The use of bio-based everyday consumable products, such as paper filters, could potentially save lives without releasing bactericides.

In the study of Gagnon J. and Calvert N. (2016), The portable water filtration system was designed and developed as a School Project at the University of Michigan Flint. This system includes a compact, lightweight water pump that filters impurities to improve drinking water quality. The water passes through a carbon element filter, an ultraviolet light, and a hollow membrane tube filter. This system can be used in developing countries where purified drinking water is scarce. The School Administrators were very astonished and accepted the project. Thus, it used different phases of product design and development processes. Engineering analyses were performed to determine head loss, sterilization time, power consumption, and stress due to impact. Most of the components were designed and developed using a 3D printer technology with ABS plastic materials. The filtration system will be

tested using Environmental Protection Agency guidelines for drinking water quality.

**Review Related Studies**

The History of Water Filters (n.d) mentions the significant contributions of scientists and medical experts in understanding the connection between microbes and disease, which has been crucial in the development of water purification methods. The discovery of the microscope by Antonie van Leeuwenhoek in the 1670s allowed for the observation of microorganisms, leading to the realization that clean-looking water is not necessarily safe. This understanding undoubtedly influenced the development of water purification technologies, including portable water purifiers, as a means to protect public health.

According to the Philippines, Water Management | Enviliance ASIA (n.d), The Republic Act No. 9275, An Act Providing for Comprehensive Water Quality Management and other Purposes more widely known as the "Philippine Clean Water Act of 2004," is the primary legislation in the Philippines that protect the country's water bodies from pollution. It was approved on March 02, 2004, and on May 16, 2005, the Department Administrative Order (DAO) 2005-10, which is its Implementing Rules and Regulations (IRR), was released. The law provides an integrated, holistic, and decentralized strategy to prevent and minimize pollution through a multi-sectoral and participatory approach involving all

stakeholders. It aims to protect, preserve, and revive the quality of fresh, brackish, and marine water from land-based sources (industries and commercial establishments, school, agriculture, and community/household activities). The Clean Water Act covers water quality management in all water bodies, reduction and control of pollution from land-based sources, and enforcement of water quality standards, regulations, and penalties. In addition, standards for the transport and disposal of effluent, sewage, and septage offsite, whether offshore or on land, as well as disposal of wastewater on land shall be formulated by DENR.

According to America's Children and the Environment (2015), The concern over contaminants in water, including microbial pathogens such as bacteria, viruses, and protozoa, has been a longstanding issue impacting public health globally. These contaminants can lead to a variety of waterborne diseases, which are a major concern worldwide. The presence of microbial contaminants like *E. coli*, *Giardia*, noroviruses, and others in water sources can lead to acute diseases such as gastrointestinal illness, developmental effects such as learning disorders, endocrine disruption, and even cancer. Children are particularly vulnerable to these contaminants due to their higher water intake relative to body weight compared to adults, making the safety of drinking water a critical issue for this demographic.

Hatt, B. E., Fletcher, T. D., & Deletić, A. (2007). A study published on PubMed discusses the treatment performance of gravel filter media in stormwater infiltration systems. The research found that gravel filters

are very effective in removing sediment and heavy metals under various water level regimes. Despite the sediment particle size distribution being much smaller than the filter media pore size, the gravel effectively trapped sediment and associated pollutants at the top of the filter. However, the study also noted that gravel filters could have been more effective at nutrient removal, particularly for dissolved nutrients.

The study and the work by Clasen et al. (2007), conducted a systematic review and meta-analysis of interventions to improve water quality for preventing diarrhea. This study highlighted the effectiveness of point-of-use water treatment methods, including chlorine disinfection, solar disinfection, and ceramic filtration, in significantly reducing diarrheal diseases, which are often caused by bacterial contamination of drinking water. The review concluded that interventions to improve the microbial quality of drinking water are effective in reducing diarrheal illness, with point-of-use disinfection showing substantial benefits. Another important study by Sobsey et al. (2008) evaluated the use of a portable, low-pressure membrane filtration device for household water treatment in rural communities. The study found that the membrane filter was highly effective in removing bacteria and viruses from drinking water, leading to a significant reduction in waterborne disease risk. The research underscored the potential of portable filtration devices to provide safe drinking water in areas without access to treated municipal water supplies.

The study of Grupper et al. (2009) explored health beliefs about bottled water among a group of participants, highlighting the role of trust and risk perceptions in influencing water consumption choices. The study found that perceptions of tap water quality were significantly influenced by trust in local water utilities and concerns about potential health risks associated with tap water consumption. This suggests that students' confidence in water sources and their risk assessments of water quality can significantly impact their preferences for bottled water over tap water or purified water.

According to Weida, K. (2019), The importance of water testing in the context of portable water purifiers cannot be overstated. And it involves analyzing water before and after purification to determine the removal efficiency of the device for various contaminants. Parameters commonly tested include microbiological contaminants (e.g., bacteria, viruses, protozoa), chemical pollutants (e.g., heavy metals, pesticides, pharmaceuticals), and physical characteristics (e.g., turbidity, color). Testing can reveal the effectiveness of the purifier and identify any potential issues that may compromise water safety. Furthermore, It is a critical component of ensuring the safety and reliability of these devices in providing clean drinking water. As portable water purification technology continues to evolve, ongoing research and development, coupled with rigorous water quality testing, will be essential for addressing the challenges of water contamination and meeting the drinking water needs of diverse populations.

**Synthesis of Related Literature and Related Study**

This section is a collective knowledge that informs and contextualizes the current study. Additionally, the researchers offer a synthesis of key findings within the literature and studies that lay the foundation for the Research.

**Synthesis of Related Literature**

The World Health Organization reports that approximately 2 billion people lack access to safe drinking water, leading to around 485,000 deaths yearly from water-related diseases. Portable water purifiers offer a solution by effectively removing contaminants and transforming dirty water into safe drinking water.

Ateneo de Manila University recognizes the importance of accessible drinking water for its community. They ensure potable water availability across campus and prioritize food safety and quality standards through their Office for Food Safety and Quality Assurance.

Ahmed et al. (2022) developed activated charcoal from walnut wastes, demonstrating significant adsorption capabilities for water purification. Alves et al. (2021) explored the effectiveness of biologically activated carbon in water treatment. Both studies highlight the efficacy of activated carbon in removing contaminants.

Rosario, Ph.D. (2012) emphasizes water's essential role in life and the Philippines' commitment to providing safe drinking water, aligning with UN Millennium Development Goals.

Punzalan C. H (2020) suggests incorporating environmental education programs in schools to raise awareness of environmental issues and promote sustainable development principles.

Wu, J., Cao, M., Tong, D., et al. (2021) discuss the use of mineral stone beads for water filtration, improving taste and quality.

Dhuria, A. (2023) highlights silica sand's effectiveness in removing impurities from water due to its high porosity and natural composition.

Dutta, S. (2022) explains how different water filtration methods affect the pH level of water and emphasizes maintaining optimal pH levels for safe drinking water.

Ottenhall et al. (2018) found that modified paper filters effectively removed bacteria from water, suggesting paper filtration as an affordable purification method.

Furthermore, Gagnon J. and Calvert N. (2016) developed a portable water filtration system for improving drinking water quality, particularly in developing countries, demonstrating the potential for innovative solutions to address water scarcity issues.

**Synthesis of Related Studies**

The History of Water Filters highlights the pivotal role of scientists in understanding microorganisms' link to disease, driving advancements in water purification methods, including portable water purifiers.

The Philippine Clean Water Act of 2004 aims to protect the country's water bodies from pollution through a comprehensive strategy involving all stakeholders and enforcing water quality standards.

Contaminants in water pose a significant public health concern globally, particularly impacting children due to their higher water intake relative to body weight. Microbial pathogens like E. coli can lead to various diseases, emphasizing the importance of safe drinking water.

A PubMed study by Hatt, Fletcher, and Deletić (2007) explores gravel filter media in stormwater systems. It reveals gravel's high efficacy in removing sediment and heavy metals despite smaller particle sizes. However, it suggests the potential for better nutrient removal, especially for dissolved nutrients.

Studies by Clasen et al. (2007) and Sobsey et al. (2008) demonstrate the effectiveness of point-of-use water treatment methods, such as chlorine disinfection and membrane filtration, in reducing diarrheal diseases and improving water quality in rural communities.

Grupper et al. (2009) found that trust in local water utilities and risk perceptions influence individuals' choices regarding water consumption, highlighting the role of perceptions in water source preferences.

Moreover, Water testing is crucial for evaluating portable water purifiers' efficiency in removing contaminants, including microbiological and chemical pollutants, ensuring safe drinking water provision.

Overall, the ongoing research and development, coupled with rigorous water quality testing, are essential for addressing water contamination challenges and meeting diverse populations' drinking water needs. These studies collectively underscore the importance of accessible and safe drinking water, as well as the efficacy of various water purification methods in achieving this goal.

# CHAPTER III

# THE PROBLEM AND ITS BACKGROUND

## Research Design

*True Experimental Research Design*; An experimental research design offers the best method to scientifically investigate if portable water purifiers improve water quality for the students of Our Lady of Lourdes College. It allows us Researchers to go beyond simply observing the situation and actively test the purifier's effects in a controlled and quantifiable manner.

## Research Locale

*Barangay Gen T. De Leon, specifically* in 5031 Gen. T. De Leon Valenzuela City, the address of Our Lady of Lourdes College. This urban setting offers population diversity, accessibility via public transportation, a range of lifestyle environments, including schools, and residential areas, proximity to local health facilities, and a strong sense of community engagement. These characteristics make it an ideal location to investigate the Water Quality of Drinkable Water, contributing valuable insights to public health research.

**Sample and Sampling Technique Used**

This Quantitative Research study (experimental) will use Simple Random Sampling. The researchers will use this sampling method to select informants randomly; there are no standard and special skills involved in using this method, which can result in a fairly reliable outcome regarding the effectiveness of portable water purifiers for drinkable water. Furthermore, the researchers will use this type of sampling to gather the required results or answers for the study.

**Instrumentation:**

Surveys and Questionnaires are used to collect data from respondents about their experiences or opinions. In this study, closed-ended questions are used to assess students' perceptions of the effectiveness of the portable water purifier. The participants can express their thoughts and experiences freely.

*Informed consent* is a crucial ethical procedure in research involving human participants, ensuring they consent voluntarily after receiving comprehensive information about the study's purpose, procedures, risks, benefits, confidentiality, and participant rights.

The *Portable water Purifier* needs to be monitored to collect data during the purification process.

*Water quality testing kits* assess the purifier's effectiveness in removing harmful microorganisms by measuring parameters like pH, turbidity, TDS, and chlorine levels in untreated and purified water. These instruments assess water quality before and after purification.

In addition to on-site water quality testing, researchers may also collect water samples for laboratory analysis. This can provide more detailed information about water contaminants and purification efficacy, complementing the findings from field tests.

**Data Gathering Procedure**

Data gathering is critical to any Quantitative Research study. It is the process of collecting data from participants to answer a research question. The specific data-gathering procedures used in a quantitative study will vary depending on the research question and the data collection type. However, some general steps are common to most quantitative data-gathering procedures:

The first step in the data collection procedure is the creation of a survey questionnaire, as it will be the primary tool for gathering data from the target sample. Before administering it to the respondents, the validators that the researchers chose will review the questionnaire to enhance its reliability and validity. Once the validators provide their feedback, the questionnaire will be modified accordingly.

After the survey questionnaire has undergone content and face validation, it is deemed suitable for dissemination to the intended

participants. During this phase, the researchers will uphold ethical principles to safeguard anonymity; the participants were given informed consent to ensure they participated voluntarily. Upon the completion of data collecting, the researchers will employ statistical methodologies to analyze and interpret the gathered data with the assistance of a qualified statistician.

**Statistical Treatment of Data**

**Weighted Mean (Average):**

Calculate the average of water before and after purification to understand the overall trend.

**Standard Deviation**:

Measure the spread of water quality measurements around the mean to understand the variability or consistency of water quality.

**T-Test**: **Two Sample Test Assuming Unequal Variance**

Compare the mean of Before and After Treatment, determine if there are significant differences between them.

**Correlation Coefficient (Pearson's r):**

Determine if there is a relationship between water quality metrics and student satisfaction ratings. A positive correlation suggests that as one variable increases, the other tends to increase as well.

OUR LADY OF LOURDES COLLEGE
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# The Effectiveness of C.L.E.A.R A.Q.U.A. Portable Purifier for Drinkable Water of Senior High School Students in Our Lady of Lourdes College

**A Quantitative Research Paper**

**Presented to the**

**Senior High School Department of**

**Our Lady of Lourdes College**

**In Partial Fulfillment of the Subject**

**Practical Research 2**

**STEM 12 - 1**

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# CHAPTER IV

## PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the results, the analysis and interpretation of data gathered from the answers to the questionnaires distributed to the field. The said data were presented in tabular form in accordance with the specific questions posited on the statement of the problem.

**1.) What are the Demographic Profile of Respondents**

**A. Age**

**B. Gender**

**C. Grade Level**

**Table 1.0A Age of Respondents**

| **SOP1.0A Demographic Profile of the Respondents** | | |
|---|---|---|
| **AGE** | **FREQUENCY** | **PERCENTAGE** |
| 16 years old | 8 | 16% |
| 17 years old | 19 | 38% |
| 18 years old | 16 | 32% |
| 19 years old and above | 7 | 14% |
| **TOTAL** | **50** | **100%** |

This table, titled "SOP1.0 A Demographic Profile of the Respondents", provides information about the age distribution of a particular group of respondents. There are a total of 50 respondents. 8 of these respondents are 16 years old, which represents 16% of the total group. 19 respondents are 17 years old, making up 38% of the

group. 16 respondents are 18 years old, representing 32% of the total. The remaining 7 respondents are 19 years old and above, making up the other 14% of the group.

In summary, the table shows that the group of respondents is primarily composed of 17 and 18-year-olds, with fewer respondents in the 16 and 19 and above age groups.

**Table 1.1B Gender of Respondents**

| **SOP1.1B Demographic Profile of the Respondents** | | |
|---|---|---|
| **GENDER** | **FREQUENCY** | **PERCENTAGE** |
| MALE | 21 | 42% |
| FEMALE | 29 | 58% |
| **TOTAL** | **50** | **100%** |

The Table 1.1 Gender of Respondents is part of the section "SOP1.1B Demographic Profile of the Respondents." It presents the gender distribution of a group of respondents. There were 50 respondents. Out of these 50 respondents, 21 were male, which represents 42 percent of the total. The remaining 29 respondents were female, accounting for 58 percent of the total.

The data illustrates the distribution of genders among a set of participants. The number of responders was 50. Among the 50 respondents, 21 were male, accounting for 42 percent of the total. Out of the total number of respondents, 29 were female, making up 58 percent of the total.

**Table 1.2C Grade Level of Respondents**

| **SOP1.2C Demographic Profile of the Respondents** | | |
|---|---|---|
| **GRADE LEVEL** | **FREQUENCY** | **PERCENTAGE** |
| GRADE 11 | 24 | 48% |
| GRADE 12 | 26 | 52% |
| **TOTAL** | **50** | **100%** |

This table, titled "SOP1.2C Demographic Profile of the Respondents", provides information about the grade level distribution of Our Lady of Lourdes College Senior High School students. There are a total of 50 respondents. 24 of these respondents are in Grade 11, which represents 48 percent of the total group. The remaining 26 respondents are in Grade 12, making up the other 52 percent of the group.

It shows that there are roughly the same number of students in Grade 11 and Grade 12 at Our Lady of Lourdes College Senior High School. Out of the 50 students surveyed, almost half 48 percent are in Grade 11, and the other half 52 percent are in Grade 12.

**2.) What is the pH Level and TDS of the C.L.E.A.R-A.Q.U.A. Portable Purifier in removing contaminants from the water in 10 days?**

**Table 2**

**The pH Level and Total Dissolved Solids of the C.L.E.A.R-A.Q.U.A. Portable Purifier in 10 Days (Before Treatment)**

| **SOP2.0 BEFORE TREATMENT** | | |
|---|---|---|
| **Day The Water Sample Was Taken** | **Total Dissolved Solids (TDS)** | **pH Level** |
| April 25, 2024 | 57 | 10.52 |
| April 29, 2024 | 62 | 10.48 |
| April 30, 2024 | 60 | 10.60 |
| May 2, 2024 | 61 | 10.58 |
| May 6, 2024 | 52 | 10.55 |
| May 7, 2024 | 55 | 10.56 |
| May 8, 2024 | 58 | 10.56 |
| May 9, 2024 | 56 | 10.49 |
| May 13, 2024 | 57 | 10.51 |
| May 14, 2024 | 57 | 10.52 |
| **AVERAGE:** | **57.5** | **10.54** |

In the data presented in SOP2 before the treatment table, it is observed that the pH levels of the water samples taken on different days ranged from 10.48 to 10.60. The Total Dissolved Solids (TDS) levels varied between 52 and 62. These values indicate that the water samples had high pH levels and TDS concentrations before treatment with the portable water purifier. The pH levels exceeding 7 suggest alkaline water, which may not be ideal for consumption due to potential health implications. Additionally, the high TDS levels indicate the presence of

various dissolved substances in the water, which could affect its taste, odor, and overall quality. The consistently high pH and TDS levels across the different sampling days highlight the need for adequate water purification methods to ensure the removal of contaminants and improve water quality. Nevertheless, this underscores the importance of using portable water purifiers to address such issues and provide safe drinking water for users.

Overall, the data from the SOP2 before treatment table emphasize the necessity of water treatment processes to enhance water quality and make it suitable for consumption, especially in environments where water contamination is a concern.

According to the World Health Organization (WHO), The pH level is an important factor in determining the safety and quality of drinking water. The pH level of 10.50 is not advisable for drinking water. The ideal range of pH levels for drinking water is generally considered to be between 6.5 and 9 pH level. Above 9 can have negative impacts on both human health and water quality. Elevated pH levels can lead to skin and eye irritations, heighten the likelihood of absorbing hazardous metals, and harm pipes and systems, hence presenting health risks and environmental concerns.

**Table 2.1**

**The pH Level and Total Dissolved Solids of the C.L.E.A.R-A.Q.U.A. Portable Purifier in 10 Days (After Treatment)**

| **SOP2.0 AFTER TREATMENT** | | |
|---|---|---|
| **Day The Water Sample Was Taken** | **Total Dissolved Solids (TDS)** | **pH Level** |
| April 25, 2024 | 57 | 9.01 |
| April 29, 2024 | 56 | 8.68 |
| April 30, 2024 | 57 | 9.11 |
| May 2, 2024 | 56 | 8.77 |
| May 6, 2024 | 52 | 8.89 |
| May 7, 2024 | 57 | 8.76 |
| May 8, 2024 | 50 | 8.51 |
| May 9, 2024 | 50 | 8.45 |
| May 13, 2024 | 50 | 8.44 |
| May 14, 2024 | 52 | 8.52 |
| **AVERAGE** | **53.7** | **8.71** |

In the data presented in SOP2 after treatment table, there is a noticeable improvement in the pH and Total Dissolved Solids (TDS) levels of the water samples after treatment with the portable water purifier. The pH levels of the water samples taken on different days ranged from 8.44 to 9.11, showing a decrease compared to the initial pH levels before treatment. The reduction in pH levels indicates that the water has moved towards a more neutral range, which is generally considered more suitable for drinking water. Additionally, the TDS levels after treatment ranged from 50 to 57, showing a decrease compared to the initial TDS levels before treatment. Moreover, the decrease in both pH and TDS levels after treatment with the portable water purifier suggests that the device effectively removed contaminants and

impurities from the water, leading to an improvement in water quality. This outcome highlights the portable water purifier's efficacy in enhancing users' safety and drinkability.

Overall, the data from the SOP2 after-treatment table demonstrate the positive impact of using the portable water purifier in reducing pH and TDS levels, thereby improving water quality and making it more suitable for consumption.

According to the review of Dutta, S. (2022), the pH level and total Dissolved Solids of filtered water depend on the type of filter used. For example, activated carbon filters can lower the pH level of water, according to the Total Dissolved Solids meter. In contrast, reverse osmosis filters do not have a significant effect on the pH level and TDS meter. In general, filtered water has a slightly lower pH level and marginally higher TDS meter than unfiltered water, but the difference is usually not significant. The pH level of filtered water is typically within the range of 6 to 9.0, and the typical range of Total Dissolved Solids Meter is within the range of 0-150, which is considered safe for drinking. Furthermore, the pH level and TDS meter of water are essential factors to consider when using a portable water purifier. While some filters are designed to raise the pH level and TDS meter of water, it is necessary to ensure that the water remains within the optimal pH range of 6 to 9.0 and the TDS meter range of 0-150, ensuring its safety and quality. Regular testing and maintenance of the water filter help ensure that the pH level and TDS meter remain within this range and that the water is safe to drink.

**3.) Does the Use of C.L.E.A.R-A.Q.U.A. Portable Purifier Significantly Reduces Bacteria in Drinking Water?**

**Table 3 Bacteria Testing**

**BGR HYDROLAB**
**TESTING CENTER**
**DOH ACCREDITATION NO. 13-005-LW-2**

**MICROBIOLOGICAL TESTING**

**REPORT OF ANALYSIS**

Name of Client : ROBERTO BRYAN CASTRO
Name of Company : OUR LADY OF LOURDES COLLEGE
Company Address : 5031 Gen. T. de Leon, Valenzuela
Contact Number/s : 0960-210-4197

**SAMPLE DESCRIPTION**

ALTAF No. : 115584
Date of Sampling : 01-May-2024
Date Received : 01-May-2024

Source of Sample: LWD – Product Water (Purified)
Time of Sampling: 02:32 PM
Sample Picked Up by : AAC

| Analysis | *Method | | **PNSDW Limit | Results | Remarks |
|---|---|---|---|---|---|
| Total Coliform | Multiple Tube Fermentation | SMEWW 9221B | <1.1 MPN / 100ml*** | <1.1 MPN / 100 ml | PASSED |
| Fecal Coliform | Multiple Tube Fermentation | SMEWW 9221E | <1.1 MPN / 100ml*** | <1.1 MPN / 100 ml | PASSED |

MPN – Most Probable Number CFU – Colony Forming Units ***for.treated water only

*The laboratory does not guarantee that the sample(s) submitted is (are) representative of the whole bulk from where it/they was (were) taken.*

**REMARKS:**

The results showed that the water sample submitted **PASSED** the DOH Standards for drinking water with respect to coliform bacteria.

**REFERENCES:**
**Philippine National Standards for Drinking Water, Department of Health, Manila, Philippines, 2007
* Standard Methods for the Examination of Drinking Water and Wastewater, 22nd edition, APHA, AWWA, WEF Washington D.C., 2012

*The results given in this report were obtained at the time of the test and refer only to the particular sample submitted. This report may not be reproduced, published or used for marketing purposes unless with written authorization from BGR Hydrolab Testing Center.*

09-May-2024
Date Released

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According to this table 3 the water sample from Our Lady of Lourdes College has passed all the microbiological tests for portable

water, meeting the Philippine National Standards for Drinking Water (PNSDW) limits for with the Total Coliform of <1.1 MPN/100 mL and with the Fecal Coliform of <1.1 MPN/100 mL. The water is deemed safe for drinking with respect to coliform bacteria.

Analyzing on this table, It complies with the limits set by the Philippine National Standards for Drinking Water (PNSDW) for Total Coliform, which is less than 1.1 Most Probable Number/100 mL, as well as for Fecal Coliform, which is also less than 1.1 Most Probable Number/100 mL from the water sample obtained from Our Lady of Lourdes College has successfully met all the microbiological tests required for potable water. The water is considered safe and drinkable in terms of coliform bacteria.

The Philippine National Standards for Drinking Water (PNSDW), established in 2007 through Administrative Order No. 2007-0012, sets strict standards for drinking water quality in the Philippines. It applies to all drinking water sources, including water refilling stations. The water sample from Our Lady of Lourdes College has passed all the microbiological tests for potable water. It meets the Philippine National Standards for Drinking Water (PNSDW) limits for both Total Coliform and Fecal Coliform, which are less than 1.1 MPN/100 mL. Therefore, the water is considered safe for drinking with respect to coliform bacteria. Moreover, it suggests that the use of portable water can significantly reduce bacteria in drinking water.

**4.) What is the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier Before and After Treatment.**

**Table 4**

**The LEVEL OF EVALUATION of Students in the Water Portable Purifier (Before Treatment)**

| **SOP4.** | | | |
|---|---|---|---|
| **What is the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier Before and After Treatment?** | **Weighted Mean** | **Verbal Interpretation** | **Rank** |
| Q1. The respondents believe that the portable water purifier is effective in removing contaminants from water. | **1.93** | DISAGREE | 2 |
| Q2. The respondents generally feel satisfied with the water quality of the portable water purifier. | **1.93** | DISAGREE | 2 |
| Q3. The respondents believe that the purified water is effective at satisfying thirst and keeping them hydrated. | **2** | DISAGREE | 5 |
| Q4. The respondents believe that the purified water appears clear before using the portable water purifier. | **1.86** | DISAGREE | 1 |
| Q5. The respondents believe that portable water purifiers are effective in removing solid particles from water. | **2.07** | DISAGREE | 6 |
| Q6. The respondents are confident that the water produced by the Portable Water purifier is safe to drink. | **2.13** | DISAGREE | 8 |
| Q7. The respondents believe that portable water purifiers are essential for maintaining water quality standards. | **1.95** | DISAGREE | 3 |

| | | | |
|---|---|---|---|
| Q8. The respondents believe that the purified water has a noticeably improved taste. | **1.97** | DISAGREE | 4 |
| Q9. The respondents are satisfied with the overall performance of the portable water purifier. | **2.08** | DISAGREE | 7 |
| Q10. The respondents consider that the Portable Water Purifier is an effective purification device to be. | **2** | DISAGREE | 5 |
| **GRAND WEIGHTED MEAN TOTAL** | **1.992** | | |

Table 4 shows the data for the perception of senior high school students at our Lady of Lourdes College before treatment. Based on the table that is categorized as "The respondents believe that the purified water appears clear after using the portable water purifier," it was ranked 1 with a weighted mean of "1.86" and an interpretation of "disagree," followed by" The respondents believe that the portable water purifier is effective in removing contaminants from water," and "The respondents generally feel satisfied with the water quality of the portable water purifier," with a weighted mean of "1.93" and an interpretation of "disagree." "The respondents believe that portable water purifiers are essential for maintaining water quality standards." has a weighted mean of "1.95" and an interpretation of "disagree." this was followed by "The respondents believe that the purified water has a noticeably improved taste." with the weighted mean of "1.97" and interpretation of "disagree." Moreover, "The respondents consider the Portable Water Purifier to be an effective purification device." and "The respondents believe that the purified water is effective at satisfying thirst and keeping them hydrated." for rank 5 with the total weighted

mean of 2 with the result of "disagree" for interpretation. Moreover, the evaluation of the portable water purifier before treatment showed that respondents were not satisfied with its performance. They disagreed that the purifier effectively removes solid particles, with a weighted mean of 2.07 and a rank of 6. Respondents were unhappy with the purifier's performance, as indicated by a weighted mean of 2.08 and a rank of 7. These results suggest that the purifier needs to be improved to meet users' needs and ensure they trust it to provide safe and clean drinking water. They also lacked confidence in the safety of the water it produced, with a weighted mean of 2.13 and a rank of 8. Most senior high school students with the data received from a total of 1.992 grand weighted mean, resulting in a disagreed interpretation.

Analyzing the evaluation of students' pre-treatment experiences with the portable water purifier reveals some unfavorable findings. The table 4 displays the statistics about the perception of senior high school students at Our Lady of Lourdes College prior to any intervention or treatment. The table categorizes the respondents' beliefs about the clarity of purified water after using a portable water purifier. It ranks this belief as 1, with a weighted mean of 1.86, indicating disagreement. The next belief is about the effectiveness of the portable water purifier in removing contaminants, which also received a weighted mean of 1.93 and an interpretation of disagreement. Lastly, the respondents generally expressed dissatisfaction with the water quality of the portable water purifier. "The respondents hold the belief that portable water purifiers are indispensable for upholding water quality standards." The weighted mean of the data is 1.95, indicating a disagreement. Subsequently, the

statement "The respondents hold the view that the purified water exhibits a significantly enhanced flavor." was presented, along by a weighted average of "1.97" and an interpretation of "disagree." In addition, the respondents perceive the Portable Water Purifier as a highly efficient purifying gadget. The respondents hold the belief that purified water is helpful in quenching thirst and maintaining hydration, as indicated by a rank of 5 and a total weighted mean of 2. The interpretation of these results is that the respondents disagree with this belief. Furthermore, the assessment of the portable water purifier prior to treatment indicated that the respondents were dissatisfied with its performance. Their contention was that the purifier does not successfully eliminate solid particles, as indicated by a weighted mean of 2.07 and a rank of 6. The respondents expressed dissatisfaction with the functioning of the purifier, as evidenced by a weighted mean of 2.08 and a rank of 7. These findings indicate that the purifier should be enhanced in order to meet the requirements of consumers and establish their confidence in its ability to deliver safe and pure drinking water. Furthermore, they expressed a lack of trust in the safety of the water it generated, as indicated by a weighted mean of 2.13 and a rank of 8. The majority of senior high school pupils obtained a grand weighted mean of 1.992, leading to a conflicting interpretation.

Murphy, H. M., Prioleau, M. D., Borchardt, M. A., & Hynds, P. D. (2010), confirm that it does need portable water purifiers, and these are efficient in eliminating impurities and meeting users' requirements for safe and pure drinking water. The devices' efficacy in preserving water quality standards and safeguarding public health leads to a substantial rise in user satisfaction and confidence. Consumers rely on portable water purifiers to ensure safe drinking water, particularly in situations involving emergency and outdoor activities. Ensuring this guarantee is crucial for the widespread adoption and utilization of these measures, as they enforce water quality rules and promote public health initiatives, hence reducing the occurrence of waterborne diseases.

**Table 4.1**

**The LEVEL OF EVALUATION of Students in the Water Portable Purifier (After Treatment)**

| **SOP4.** | | | |
|---|---|---|---|
| **What is the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier Before and After Treatment?** | **Weighted Mean** | **Verbal Interpretation** | **Rank** |
| Q1. The respondents believe that the portable water purifier is effective in removing contaminants from water. | **3.38** | STRONGLY AGREE | 4 |
| Q2. The respondents generally feel satisfied with the water quality of the portable water purifier. | **3.4** | STRONGLY AGREE | 3 |
| Q3. The respondents believe that the purified water is effective at satisfying thirst and keeping them hydrated. | **3.4** | STRONGLY AGREE | 3 |
| Q4. The respondents believe that the purified water appears clear after using the portable water purifier. | **3.38** | STRONGLY AGREE | 4 |
| Q5. The respondents believe that portable water purifiers are effective in removing solid particles from water. | **3.55** | STRONGLY AGREE | 1 |
| Q6. The respondents are confident that the water produced by the Portable Water purifier is safe to drink. | **3.45** | STRONGLY AGREE | 2 |

| | | | |
|---|---|---|---|
| Q7. The respondents believe that portable water purifiers are essential for maintaining water quality standards. | **3.35** | STRONGLY AGREE | 5 |
| Q8. The respondents believe that, the purified water has a noticeably improved taste. | **3.38** | STRONGLY AGREE | 4 |
| Q9. The respondents are satisfied with the overall performance of the portable water purifier. | **3.35** | STRONGLY AGREE | 5 |
| Q10. The respondents consider that the Portable Water Purifier is an effective purification device to be. | **3.55** | STRONGLY AGREE | 4 |
| **GRAND WEIGHTED MEAN TOTAL** | **3.419** | | |

According to Table 4.1, it shows the data on the level of evaluation of students in the water portable purifier after treatment, which the respondents answered. Based on the table that is categorized as "The respondents believe that the portable water purifier is effective in removing contaminants from water." a total of 3.38 weighted mean results in a strongly agreed interpretation. Subsequently followed, the categorization of "The respondents generally feel satisfied with the water quality of the portable water purifier." has a total of 3.4 weighted mean resulting in a strongly agreed interpretation. Furthermore, the categorization of "The respondents believe that the purified water is effective at satisfying thirst and keeping them hydrated." has a 3.4 weighted mean, resulting in a strongly agreed interpretation. This was followed by the categorization of "The respondents believe that the

purified water appears clear after using the portable water purifier." which has a total of 3.38 weighted mean, which results in a strongly agreed interpretation. It was also followed by the categorization of "The respondents believe that portable water purifiers are effective in removing solid particles from water." It has a 3.55 weighted mean, resulting in a strongly agreed interpretation. One was also followed by the categorization of "The respondents are confident that the water produced by the Portable Water purifier is safe to drink." this has a total of 3.45 weighted mean, which results in a strongly agreed interpretation. This is strictly followed by the categorization of "The respondents believe that portable water purifiers are essential for maintaining water quality standards." It has a 3.35 weighted mean, resulting in a strongly agreed interpretation. Followed by the categorization of "The respondents believe that, the purified water has a noticeably improved taste." which has a total of 3.38 weighted mean, which results in a strongly agreed interpretation. Next, the categorization of "The respondents are satisfied with the overall performance of the portable water purifier." has a total of 3.35 weighted mean, which results in a strongly agreed interpretation. Additionally, 3.55 weighted mean results in a strongly agreed interpretation as "The respondents consider that the Portable Water Purifier is an effective purification device to be." Lastly, it has a total of 3.42 Grand weighted mean that is equivalent to Strongly Agree Verbal Interpretation.

Analyzing the assessment of students' post-treatment experiences with the Water Portable Purifier is quite positive, as shown in Table 4.1. The data, which is shown as weighted mean scores, shows that there is broad agreement among the pupils regarding the purifier's efficacy. The student's high level of trust in the purifier's purifying process is reflected in their weighted mean of 3.38 for agreement on the device's ability to remove pollutants. Furthermore, a weighted mean of 3.4 indicates that students are generally satisfied with the quality of the water, indicating a positive change in their opinion of the filtered water. Beyond providing pure water, the students also overwhelmingly agree—again with a weighted mean of 3.4—that the purifier is effective at slaking thirst and keeping bodies hydrated. This emphasizes the practical benefits of utilizing the equipment. Also, the students overwhelmingly agree that the water looks clear after purification. Therefore, the weighted mean of 3.38 for clarity matches their expectations. Overall, the students' answers demonstrate a significant improvement in their perceptions of the portable water purifier's ability to provide drinkable water that not only satisfies their standards for purity but also helps them meet their hydration requirements. The results shown in Table 4.1 prove the purifier's ability to provide pure drinking water. In addition, it has a 3.1419 Grand weighted mean that is equivalent to Strongly Agree Verbal Interpretation.

In the study of Wilson, C., Hill, V., & Phillips, R. (2012), it supports the view that portable water purifiers are effective in removing contaminants and satisfying users' needs for safe, clean drinking water. High user satisfaction and confidence in these devices underscore their importance in maintaining water quality standards and ensuring public health. Moreover, users highly rate the performance of portable water purifiers. In various studies, respondents expressed strong agreement regarding the devices' effectiveness and reliability. These positive perceptions are critical for the continued development and improvement of water purification technologies.

**5.) Is there a significant difference between the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A Portable Purifier Before and After Treatment?**

**Table 5**

**Hypothesis Testing in Comparing the Level of Evaluation of Students in C.L.E.A.R-A.Q.U.A. Portable Purifier Before and After Treatment**

| **t-Test: Two-Sample Assuming Unequal Variances** | | | | |
|---|---|---|---|---|
| | | | | |
| | *BEFORE Treatment* | *AFTER Treatment* | | |
| Mean | 18.04 | 33.1 | | |
| Variance | 63.10041 | 52.90816 | | |
| Observations | 50 | 50 | | |
| Hypothesized Mean Difference | 0 | | | |
| df | 97 | | | |
| t Stat | 9.88701 | | | |
| P(T<=t) one-tail | 1.18E-16 | | | |
| t Critical one-tail | 1.660715 | | | |
| P(T<=t) two-tail | 2.35E-16 | | | |
| t Critical two-tail | 1.984723 | | | |

| t-Test: Two-Sample Assuming Unequal Variances | | | | |
|---|---|---|---|---|
| | | | | |
| | *BEFORE* | *er treatment* | | |
| Mean | 18.04 | 33.1 | | |
| Variance | 63.10041 | 52.90816 | | |
| Observati | 50 | 50 | | |
| Hypothesi | 0 | | | |
| df | 97 | | | |
| t Stat | -9.88701 | | | |
| P(T<=t) on | 1.18E-16 | | | |
| t Critical o | 1.660715 | | | |
| P(T<=t) tw | 2.35E-16 | | | |
| t Critical t | 1.984723 | | | |

According to this table, the results are finally revealed upon analyzing the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier between Before and After Treatment. We can conclude in our study that the t-statistic of 8.94 is significantly different from the critical t-statistic of 1.98, as the two-tailed test is being conducted. Consequently, there is a statistically significant disparity in the average Evaluation scores between individuals of Senior High School Students at Our Lady of Lourdes College. The extremely low p-value provides additional evidence in favor of this conclusion, indicating that the likelihood of obtaining such data if there were no significant differences in the Level of Evaluation of Students in C.L.E.A.R-A.Q.U.A. Portable Purifier Before and After Treatment is extremely small.

Analyzing this table, the results are now disclosed after assessing the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier before and after treatment. Based on our analysis, we can confidently state that the t-statistic of 8.94 is significantly different from the critical t-statistic of 1.98, given that we are conducting a two-tailed test. Therefore, there is a notable and meaningful difference in the average Evaluation scores among Senior High School Students at Our Lady of Lourdes College. The exceptionally small p-value provides further evidence supporting this conclusion, stating that the Portable Water Purifier is effective, and thus suggesting that the probability of obtaining such data in the absence of significant differences in the Level of Evaluation of Students in C.L.E.A.R-A.Q.U.A. Portable Purifier Before and After Treatment is truly low.

In the study of Quasi Ibrahim et al. (2022), supports the effectiveness of portable water purifiers, emphasizing their role in water treatment. Studies analyze THM formation rates, treatment plant efficiency, and the removal of contaminants using adsorbents. It also highlights the efficiency of water filters in removing turbidity, odor, and microbial content from lake water. Additionally, a review of wastewater treatment techniques underscores the effectiveness of adsorption methods in water purification. These sources collectively demonstrate the efficacy of portable water purifiers in enhancing water quality and safety.

# CHAPTER V

# SUMMARY OF FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

In this chapter, our mixed-approach analysis focuses on answering fundamental questions that show the effectiveness of C.L.E.A.R. A.Q.U.A. Portable Purifier. By delving into data, we reveal insights such as the pH level, TDSS level, and MPN Bacteria Testing, ensuring its Water Quality. Additionally, we explore the demographic profiles of our respondents, offering valuable context to the observed purified potable water.

Finally, we found a significant difference in the evaluation levels of students before and after treatment with the C.L.E.A.R-A.Q.U.A. Portable Purifier The numerical data enables us to draw tangible conclusions, empowering future efforts for the effectiveness of the C.L.E.A.R-A.Q.U.A. Portable Purifier.

## Summary of Findings

The purpose of this study was to determine the Effectiveness of C.L.E.A.R A.Q.U.A. Portable Purifier for Senior High School Students of Our Lady of Lourdes College. The researchers employed an Experimental Research design as their methodology. A Face-to-Face Survey using a 4-point Likert scale was utilized to collect the data. The investigation for these studies was carried out in the first semester of the 2023–2024 academic year.

The following findings are the answers to the following research questions:

**Statement of the Problem**

**Research Question #1: What are the Demographic Profile of Students?**

**a) Age**

**b) Gender**

**c) Grade Level**

The findings indicate that 58% of the respondents are female, while the other 42% are male. The respondents grade levels range from Grade 11 48%to Grade 12 52%, while the age of the students ranges from 17 years old 38%, followed by 18 years old 32%, and then 16 years old 16%, and for the last range of respondents, 19 years old and above, with 14%

**Research Question #2: What are the pH Level and Total Dissolved Solids of the C.L.E.A.R-A.Q.U.A. Portable Purifier in removing contaminants from the water in 10 days?**

In **Before Treatment**, the average pH level was 10.54, and the average Total Dissolved Solids (TDS) was 57.5. These high values indicated the presence of various dissolved substances in the water, affecting its taste, odor, and overall quality. The high pH level suggested that the water was alkaline, which may not be ideal for consumption due to potential health implications. However, **After Treatment**, the water purifier effectively reduced the pH level to an average of 8.71 and the TDS to an average of 53.7. The reduction in both pH and TDS levels suggests that the purifier effectively removed contaminants and impurities from the water, leading to an improvement in water quality.

**Research Question #3: Does the Use of C.L.E.A.R-A.Q.U.A. Portable Purifier Significantly Reduces Bacteria in Drinking Water?**

The water sample collected from Our Lady of Lourdes College complies with the Philippine National Standards for Drinking Water (PNSDW) upon successfully passing all microbiological tests. The results suggest that the water is safe for consumption as the levels of both total and fecal coliform bacteria are below the threshold of 1.1 MPN/100 mL. ADMINISTRATIVE ORDER No. 2007-0012, issued in 2007, establishes the Philippine National Standards for Drinking Water. It assesses the water's fitness for human consumption and highlights the effectiveness of portable water purifiers in reducing bacterial contamination in drinking water.

**Research Question #4 What is the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier Before and After Treatment.**

Our investigation extends to identifying the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier Before and After Treatment and contributing essential knowledge to Our Lady of Lourdes College. The Before Treatment received a total of **1.99**, resulting in a disagreed interpretation, while the After Treatment received a **3.42** grand weighted mean, resulting in a strongly agreed interpretation.

**Research Question #5 Is there a significant difference between the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier Before and After Treatment?**

The outcomes become evident once a comprehensive analysis compares the Level of Evaluation of Students in the C.L.E.A.R. A.Q.U.A. Portable Purifier Before and After Treatment Based on our research findings, the t-statistic of 9.89 demonstrates a significant difference compared to the two-tailed critical t-statistic of 1.98, given a two-tailed test. Hence, a noteworthy disparity in the average scores of the Level of Evaluation of Students can be observed between Before and After Treatment based on statistical measurements. The existence of a small p-value strengthens the support for this assertion by suggesting that the probability of obtaining such data under the assumption that there are no noteworthy distinctions in the Level of Evaluation of Students between Before and After Treatment is highly improbable.

**Conclusion**

The researchers carefully interpreted the presentation of the data and came up with the following conclusions on this study that set its basis according to its findings:

1. Our research uncovered significant findings regarding the demographic profile of the respondents. The demographic profile helped provide context to the evaluation of the C.L.E.A.R. A.Q.U.A. Portable Purifier and its effectiveness among senior high school students. Thus, understanding the age, gender, and grade level distribution of the

respondents allowed for a more comprehensive analysis of the study results and their implications.

2. The C.L.E.A.R. A.Q.U.A. Portable Purifier significantly improved water quality at Our Lady of Lourdes College. From a 10.54 pH level and 57.5 TDS, immediately reduce the water's pH to 8.71 and TDS to 53.7, reducing contaminants and impurities and making it safer and more suitable for consumption.

3. Our Lady of Lourdes College's water sample meets the Philippine National Standards for Drinking Water (PNSDW), with total and fecal coliform levels below 1.1 MPN/100 mL, confirming its safety and highlighting the effectiveness of portable water purifiers.

4. The study at Our Lady of Lourdes College revealed a significant improvement in students' evaluation of the C.L.E.A.R. A.Q.U.A. portable purifier. Previously, the purifier had a low evaluation score of 1.93, but after Treatment, it had a high score of 3.42, indicating a significant improvement in its effectiveness and efficiency.

5. Furthermore, the hypothesis testing confirmed that there is a statistically significant difference between the student's evaluation of the C.L.E.A.R. A.Q.U.A. Portable Purifier before and after Treatment. The t-statistic of 9.89, compared to the critical value of 1.98, indicates a substantial difference in evaluations before and after Treatment. The small p-value further supports this finding, suggesting that the observed improvement is unlikely to be due to chance. The students' evaluations significantly enhance the purifier's effectiveness after-treatment.

**Recommendation**

The study on the effectiveness of C.L.E.A.R-A.Q.UA. Portable Purifier for drinkable water among senior high school students in Our Lady of Lourdes College highlights the importance of addressing the issue of safe drinking water in educational settings. Given the mixed evidence on the effectiveness of the pilot program in providing safe drinking water in schools, it is crucial to consider the following recommendations to improve the sustainability and impact of such initiatives:

**1.) Methodology Improvement** – Conduct a more detailed assessment of the water quality before and after the intervention to better understand the purifiers' effectiveness in removing contaminants. Incorporate a control group to compare the intervention's outcomes with a non-intervention group to isolate the purifiers' impact. Furthermore, the sample size should be increased to ensure more representative results and to account for potential variations in water quality and usage patterns.

**2.) Sustainability and Maintenance** – Developing a plan for the long-term maintenance and replacement of consumables, including a budget and a system for monitoring and reporting on usage and maintenance. Moreover, a system for training and supporting teachers and students on the proper use and maintenance of the purifiers should be established. However, partnering with local organizations or government agencies should be considered to ensure the sustainability of the program beyond the initial intervention.

**3.) Communication and Awareness** - Develop a comprehensive communication strategy to raise awareness about the importance of safe drinking water and the benefits of the purifiers among students, teachers, and parents. Additionally, utilizing the multiple channels, including social media, school announcements, and community events, to reach a broader audience. Lastly, incorporating educational materials and activities into the curriculum to promote water safety and hygiene practices.

**4.) Monitoring and Evaluation** - Establishing a system for regularly monitoring and evaluating the effectiveness of the purifiers and the overall program. Collect data on water quality, usage patterns, and maintenance to identify areas for improvement and track progress over time. Use the data to adjust the program as needed and to inform future interventions.

**5.) Collaboration and Partnerships Foster partnerships with local organizations, government agencies, and other stakeholders to leverage resources, expertise, and funding for the program** - Collaborate with other schools and educational institutions to share best practices and scale up the program.

**6.) Lessen the use of Plastic Waste and Ensuring Safeness** - Determine if the portable water purifier can provide safe, portable water to students, reducing their reliance on plastic bottled water and minimizing plastic waste. Lastly, assessing the impact of the purifier on students' health, tracking any reduction in water-related diseases and illnesses.